



**FACULTY OF COMPUTER SCIENCE AND MATHEMATICS
BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)
WITH HONOURS**

SESSION II 2025/2026

**CSE 3433 – SOFTWARE ARCHITECTURE
INDIVIDUAL LOGBOOK**

**PROJECT TITLE: SERVICE-ORIENTED ARCHITECTURE / SMART CAMPUS
SYSTEM**

GROUP NUMBER: G11

PREPARED FOR:

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PREPARED BY:

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SUBMITTED BY:

19 JUNE 2026

1.0 Team Information

NAME	MATRIC NO	ROLE
SARAH AMIRA HERI	S75812	Project Manager & System Analyst
ELSA FADLIN BINTI MERSING	S75412	Frontend Developer & Tester
SITI AISYAH ARWIAH BINTI YUNDI	S74256	Backend Developer & Database Designer

2.0 Weekly Progress Report

WEEK 4

Item	Description
Week	Week 4 (2 – 8 March 2026)
Weekly Activities	Brainstormed the foundational service-oriented architecture (SOA) components for the Smart Campus Support System.
Individual Contributions	Conducted requirements gathering for the Complaint Module. Made the first draft for the core complaint submission and ticketing schemas..
Challenges	Determining the loose coupling between the backend database schema and autonomous service modules.
Action Plans	Conducted research regarding loose coupling between services and isolated database boundaries in SOA

WEEK 5

Item	Description
Week	Week 5 (9 – 15 March 2026)
Weekly Activities	Drafted the first deliverable project proposal report for submission on 16th April 2026..
Individual Contributions	Focused on defining the database constraints, entity relations, and architectural style justifications from a backend perspective.
Challenges	Managing backend service endpoints and database schemas before frontend interface designs are officially established.
Action Plans	Review with lecturer on the backend architecture proposed in the proposal.

WEEK 6

Item	Description
Week	Week 6 (16 – 22 March 2026)
Weekly Activities	Finalized the project proposal report.
Individual Contributions	Contributed to System Decomposition by structuring a Component Table defining database tables, entities, and REST API data structures for the Complaint Module.
Challenges	Ensuring clear data boundaries to split user-facing data

	presentation from foundational backend transaction tables.
Action Plans	Collaborate with the group's Systems Analyst to explicitly frame Boundary Statements for each autonomous sub-service block.

WEEK 7

Item	Description
Week	Week 7 (23 – 29 March 2026)
Weekly Activities	Designed the preliminary SOA architecture.
Individual Contributions	Developed a high-level database architecture diagram showing entity relationships and data-flow interactions between backend.
Challenges	Defining efficient data communication mechanisms and API routes between backend services.
Action Plans	Research backend service integration techniques, CRUD routing, and data payload optimization approaches.

WEEK 8

Item	Description
Week	Week 8 (30 March – 5 April 2026)
Weekly Activities	Evaluated internal SOA structural alternatives and finalized the specific implementation approach for our mandated SOA system.
Individual Contributions	Reviewed core SOA data-handling design principles and refined the entity-relationship mapping, ensuring the database triggers aligned with our chosen RESTful API-driven SOA sub-pattern.
Challenges	Deciding whether to route backend service requests through a shared database schema or completely isolated data tables while still adhering to the required SOA loose-coupling principles.
Action Plans	Document database design justifications, backend service boundaries, and internal architectural design decisions within our mandated framework.

WEEK 9

Item	Description
Week	Week 9 (6 – 12 April 2026)
Weekly Activities	Developed UML diagrams for the architecture design.
Individual Contributions	Created the backend perspective of the System Context Diagram, detailed Component Diagram for data layers, and Interaction Diagrams for Complaint workflows.
Challenges	Maintaining data integrity and exact matching between UML diagrams and database schema requirements.
Action Plans	Validate UML diagrams and relational schemas with team

	members and supervisor.
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WEEK 10

Item	Description
Week	Week 10 (13 – 19 April 2026)
Weekly Activities	Designed database structure and service integration workflow.
Individual Contributions	Defined strict database entities, foreign keys, relationships, indexation rules, and backend service-to-service data flows for complaint logging.
Challenges	Ensuring backend services can query and exchange relational data efficiently while remaining loosely coupled.
Action Plans	Prepare final database schema design, SQL DDL scripts, and backend integration strategy documentation.

WEEK 11

Item	Description
Week	Week 11 (20 – 26 April 2026)
Weekly Activities	Prepared prototype implementation plan.
Individual Contributions	Documented database connectivity structures, backend folder modules, service access objects (DAOs), and implementation scopes for the Complaint Module.
Challenges	Translating theoretical data schemas and architecture patterns into a practical, implementable codebase.
Action Plans	Begin developing backend system modules and setting up SQL environments according to SOA design

WEEK 12

Item	Description
Week	Week 12 (27 April – 3 May 2026)
Weekly Activities	Implemented system prototype and service components.
Individual Contributions	Assisted in developing server-side logic, routing endpoints, Java beans, and database connectivity modules via JDBC/Hibernate..
Challenges	Integrating frontend interfaces seamlessly with backend endpoints and complex database operations.
Action Plans	Conduct backend unit testing, check database connectivity, and fix identified query optimization issues.

WEEK 13

Item	Description
Week	Week 13 (4 – 10 May 2026)
Weekly Activities	Finished Project Observation 2 and compiled all layout assets for the group submission report.
Individual Contributions	Generated backend system interaction logs, database ERD screenshots, repository file maps, and API documentation for inclusion in the final report.
Challenges	Formatting and integrating database schema descriptions and technical backend architectures cleanly into a single unified layout.
Action Plans	Complete final backend reviews, inspect constraints, and assemble the individual logbook.

3.0 Gen AI Usage Record

Week	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	ChatGPT	Understand software architecture concepts	"What are the fundamental principles of software architecture?"	Identified architecture principles, quality attributes, and architectural styles.	Asked for explanations of core architecture concepts and principles.
Week 3	ChatGPT	Explore Architecture Patterns	"What architecture patterns are commonly used in software development?"	Identified Layered, SOA, Microservices, and Event-Driven Architecture patterns.	Requested comparisons between common architecture patterns.
Week 3	Gemini	Database Architectural Paradigms	"Should a Service-Oriented Architecture (SOA) use a shared database pattern or a database-per-service pattern?"	Evaluated data isolation, data integrity risks, and loose coupling benefits in SOA.	Selected a hybrid approach to maintain data boundaries between distinct campus modules.
Week 3	Gemini	Backend Framework Evaluation	"Which backend framework is best suited for building an enterprise SOA system using Java: Spring Boot or Jakarta EE?"	Provided a comparative analysis of dependency injection, runtime weight, and integration ease.	Decided on a lightweight setup aligned with the project's technical scope.
Week 4	Gemini	Requirement Analysis	"Can you help me explain where the non-functional requirements	Produced a structured explanation requirements list.	Used AI to determine system requirements.

			in the system?"		
Week 4	Gemini	Service Identification	"How can I divide my system into separate services using SOA?"	Identified candidate services and responsibilities.	Determined possible services in the system.
Week 4	ChatGPT	Complaint Module Schema Drafting	"What are the essential database attributes required for a campus complaint logging and tracking module?"	Generated a list of attributes including ticket ID, complainant ID, status states, and timestamps.	Crafted the initial relational layout for the Complaint Module database tables.
Week 4	Gemini	API Endpoint Planning	"How should I structure the REST API endpoints for a multi-tenant backend complaint system?"	Provided a set of standardized endpoint structures (POST, GET, PUT) for handling tickets.	Structured the interface routing maps for service-to-service communication.
Week 5	ChatGPT	Learn SOA Principles	"Can you explain the main principles of Service-Oriented Architecture (SOA)?"	Learned loose coupling, reusability, autonomy, and interoperability.	Requested explanations of SOA concepts.
Week 5	Gemini	Service Boundary Design	"How do I determine the boundaries between services in an SOA system?"	Established clearer service responsibilities.	Defined service responsibilities.
Week 6	Gemini	Architectural Design	Can you suggest an SOA architecture	Produced a high-level architecture structure	Defined service responsibilities

			design for my system?"		
Week 6	Gemini	Quality Attribute Analysis	"How do scalability, security, and reliability affect SOA design?"	Identified important quality attributes.	Examined system quality requirements.
Week 6		Database Indexing for High Performance	"Which columns in a high-volume complaint ticketing system should be indexed to optimize fetch query times?"	Recommended indexes on status enums, foreign user keys, and timestamp composite keys.	Applied indexing optimization strategies to the system's database schema draft.
Week 7	Gemini	Service Communication	"How do services communicate with each other in SOA?"	Understood synchronous and asynchronous communication.	Learned how services communicate.
Week 7	ChatGPT	REST API vs. Message Broker	"For a complaint module, should new complaint submissions use REST API or a message queue like RabbitMQ to inform administrators?"	Outlined tradeoffs between immediate synchronous responses (REST) and asynchronous queue durability (RabbitMQ).	Designed a hybrid approach where user submissions use REST but backend notification dispatches are offloaded to an async worker queue.
Week 8	ChatGPT	Architecture Selection	"Since we are mandated to use SOA for our campus system, which specific SOA deployment pattern (e.g., RESTful web	Recommended a RESTful, API-driven SOA pattern over heavy ESB middleware for a lightweight, responsive campus system.	Selected and justified the internal architectural sub-patterns within our mandated SOA framework.

			services vs. message-driven) is most suitable for our project requirements and why?"		
Week 8	Gemini	UML Diagram Support	"How do UML diagrams should I include in my software architecture project?"	Identified Use Case, Class, and Sequence Diagrams.	Created architecture-related diagrams.
Week 9	ChatGPT	Dynamic Modelling	"Can you help me create a sequence diagram for a backend complaint processing module?"	Produced interaction flow between components.	Developed system interaction flows.
Week 9	ChatGPT	Business Perspective Analysis	"Can you explain the revenue streams for my software system?"	Generated descriptions for revenue models.	Explored revenue opportunities.
Week 10	Gemini	Partnership Analysis	Who should be the key partners for my software system?"	Identified cloud, payment, and API providers.	Identified external stakeholders.
Week 10	ChatGPT	API Planning	"What APIs should I integrate into my system?"	Recommended payment, location, and notification APIs.	Planned external integrations.
Week 10	Gemini	Data Architecture Support	"How should my database entities be distributed across services	Suggested data ownership and service-specific data boundaries.	Used AI to review database and service relationships.

			in SOA?"		
Week 11	Gemini	Documentation Support	"Can you help me explain the fundamental principles of software architecture for my report?"	Produced report-ready explanations.	Improved report content.
Week 11	ChatGPT	Prototype Planning	"What should I write in the prototype implementation section?"	Generated prototype documentation content.	Defined implementation scope
Week 12	ChatGPT	Development Guidance	"How should I structure my Java/JSP application database connection layer for this project?"	Produced implementation guidance and architecture alignment.	Assisted during implementation.
Week 12	Gemini	Database Integration	"How do I connect my database in JSP using JDBC safely?"	Established database connectivity approach.	Connected application components securely.
Week 12	ChatGPT	Prototype Evaluation	"Does my database implementation reflect the alignment between design and implementation?"	Produced recommendations for alignment between design and implementation.	Used AI to assess implementation progress.
Week 13	Gemini	Testing Support	"Can you help me create backend test cases and SQL query checks for my system?"	Produced functional testing scenarios.	Prepared validation activities.

